



SOMAIYA
VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

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Abstract

Efficient power system management depends on accurate electricity load analysis, particularly in weather-sensitive environments. While traditional machine learning models achieve high predictive accuracy, they often lack the interpretability required for reliable real-world decision-making. This study proposes an integrated framework that bridges this gap by combining Explainable Artificial Intelligence (XAI), Causal Inference, and traditional statistical modeling to analyze the impact of weather variables specifically temperature and relative humidity on power demand.

To ensure both global and local interpretability, the framework utilizes SHAP and LIME techniques. Moving beyond simple correlation, the system incorporates the EconML library, employing Double Machine Learning (DML) and Causal Forests to estimate Average Treatment Effects (ATE) and Heterogeneous Treatment Effects (HTE). This approach allows for a granular understanding of how weather impacts vary across different seasonal and temporal conditions. A Multiple Linear Regression (MLR) model serves as a baseline for performance comparison.

The models were trained and validated using two annual cycles of time-series data. Results indicate that the EconML-based causal framework significantly outperforms the traditional MLR baseline. The causal model achieved a Mean Absolute Percentage Error (MAPE) of 1–1.2%, a substantial improvement over the 12–16% MAPE produced by the MLR. Findings confirm that temperature is the primary driver of load variations, with humidity playing a secondary role. By integrating causal reasoning with XAI, this framework provides a transparent, scalable, and highly accurate solution for future smart grid applications and energy management systems, offering deeper insights into the underlying drivers of demand behavior.

Keywords Electricity · Load Forecasting · Causal Inference · Explainable AI · SHAP · EconML Double Machine Learning